

UNIVERSITY OF BRISTOL
Summer Reassessment for Fall 2024
Topics in CS Question

SEMT30003/4
Computational Neuroscience

TIME ALLOWED:
1 hour

This paper contains one in-depth question with three sections that contain multiple parts. **All** components must be attempted and will be used for assessment. The maximum for this section is 50 marks.

Word counts are not mandatory, but rather hints to help you avoid spending more time than needed. You may exceed the advised word count if you wish. You may also use diagrams and mathematics as appropriate in short answer questions.

Please write your student number and/or username on the line below.
(To ensure blinded marking, do not write your name)

Other instructions:

- Only approved non-programmable calculators are permitted
- Write your answers in the question paper
- Blank pages are provided at the end if you need extra space. Clearly label which answer you are writing.

(a) [6 marks] General knowledge: (3–9 minutes, 1–3 minutes per question)

- i. Discuss rate and spike-timing codes in the context of calculation of inter-aural time delays by the brainstem, and in continuous modulation and control of muscle dynamics by motor neurons. Which scenario is better captured by rate coding? What are the key differences between them? (≈ 50 –100 words) [2]

When calculating inter-aural time delays, individual neurons compare the precise timing of individual spikes. This is more consistent with a spike timing code. When controlling muscles, the low-pass nature of the musculoskeletal plant mean that precise spike timings matter little; The essence of rate coding is that the exact timing and identity of spikes is unimportant, and we can average over time or neurons. Rate coding would be appropriate for modelling motor output.

Addendum: Markers should consider “rate” and “frequency” interchangeable here. The distinction is not necessary for this question, but for those interested: The word frequency is sometimes used when neural spiking is roughly periodic (not homogeneous Poisson or random). The standard Wilson-Cowan-Amari neural field reduction used to justify rate models typically thinks in terms of the average fraction of a neural population that is active, and this does not require periodicity. However, the f – I curve for the LIF model is an example of a rate reduction where the rate truly is a frequency of periodic spiking. There are also other reductions, such as the Byrne-Coombes model, that derive a continuous rate-like quantity from a population of periodically spiking neurons.

Students do not need to use the jargon “low-pass” or “musculoskeletal plant” for full marks, any equivalent statement in more elementary or accessible language is fine.

- ii. Hubel and Wiesel’s recordings from primary visual cortex (V1, Brodmann area 17) found visually tuned neurons they termed ‘*simple cells*’. What types of visual input do simple cell’s respond to, and how do they differ from the so-called ‘*complex cells*’? (≈ 50 –100 words) [2]

Simple cells respond to oriented edges (or “gratings”). Unlike complex cells, they do not exhibit spatial (also accepted: phase) invariance; Simple cells care about the specific patterns of light/dark and specific location. A simple cell might, for example, fire for a dark-to-light edge in part of the visual field, but not fire for a light-to-dark edge.

- iii. Most retinal ganglion cells do not transmit absolute luminance or colour information. Instead, they spike less often and only when they detect changes over time, space, or a mixture of the two. Broadly, what is the name for this type of encoding and what are its advantages? (≈ 50 –100 words) [2]

“Sparse coding” or “sparsity”; Two things are happening here: Firstly, by transmitting only differences or changes, the retina is performing lossy compression to discard unimportant information. This reduces the overall amount of information that needs to be communicated. Secondly, sparse coding uses slightly richer per-neuron tuning to allow somewhat lower per-neuron activity levels (which may have metabolic advantages). Combined, these reduce bandwidth costs to efficiently compress and transmit visual information through the optic nerve.

Addendum: The scientific consensus and prevailing terminology in the field is vague/controversial enough to allow wiggle room here. For full marks, students should say the word “sparse”, and clearly describe one key feature of sparse coding or the retinal code, without adding any claims or information that is scientifically incorrect. Yes, retinal coding is sparse, but is it more sparse than average for a neural code? If not, then is it right to emphasise the sparsity itself as the defining feature? Normative motivations for sparsity also vary: Principally, it is the *compression* that is key for fitting visual data into the bandwidth of the optic nerve. But, other design choices, like restricting high-resolution image forming to the fovea, help immensely, and these are unrelated to sparse coding. Yes, sparsity may have metabolic advantages, and yes, it may be a crucial aspect of codes that communicate varying information rates along the fixed-bandwidth of the optic nerve (see Rule, Sorbaro, Hennig 2020), but it is unclear whether sparsity can be truly said to arise from or serve any specific purpose. For example, it may be the case that the type of feature extraction and lossy compression is the best that can be achieved with local connectivity and limited resources, and the resulting sparsification is a side-effect of being unable to use e.g. global or strongly recurrent dynamics to achieve a denser code using fewer axons. Or, it may be that the sparse code provides downstream computational benefits, and has little to do with “efficiency” (in an information theoretic, metabolic, or wiring cost sense). Mentioning the representational or computational benefits of sparsity is also valid here, although in lecture these were largely discussed in the context of area CA3.

(b) [22 marks] Leaky integrate-and-fire dynamics: (10–20 minutes)

Consider a Leaky Integrate-and-Fire (LIF) neuron obeying the following dynamics:

$$RC\dot{v} = E_\ell - v + IR$$

(1)

if $v(t) > v_{\text{threshold}}$ then $v(t) \leftarrow v_{\text{reset}}$ and emit a spike

Assume that $v_{\text{reset}} \leq E_\ell < v_{\text{threshold}}$.

- i. Give the physiological interpretation of C , R , E_ℓ , I , $v_{\text{threshold}}$, and v_{reset} . Provide valid scientific units for each quantity, and discuss how this model approximates the ionic currents across the neuronal membrane during the action potential. (address all 5 variables ≈ 50 – 200 words) [6]

C : (farads or farads per area or equivalent) Membrane capacitance, relationship between charge flow across membrane and change in membrane voltage, measures stored electrostatic field;

R : (Ohms, Siemens, Siemens per area, or equivalent) The membrane resistance determines how quickly the neuron is pulled toward its resting potential. Smaller values mean the membrane is more leaky.

E_ℓ : (volts, or equivalent) Resting potential, the effective reversal potential of all channels contributing to the leak conductance; Determine the baseline membrane voltage without input

I : (amperes, or equivalent) Injected or applied current, or current arising from other ion channels.

$v_{\text{threshold}}$: (volts, or equivalent) The voltage above which positive feedback will trigger an action potential. Often set to -55 mv.

v_{reset} : (volts, or equivalent) The voltage the neuron returns to after spiking. This is usually at or below the resting potential. This reflects the typical re-polarised or hyper-polarised membrane voltage reached during the repolarisation phase of the action potential, due to the outflow of positive potassium ions from the cell.

- ii. Show that Equation (2) (below) is the solution to the ODE in Equation (1) if the spiking threshold and reset are ignored:

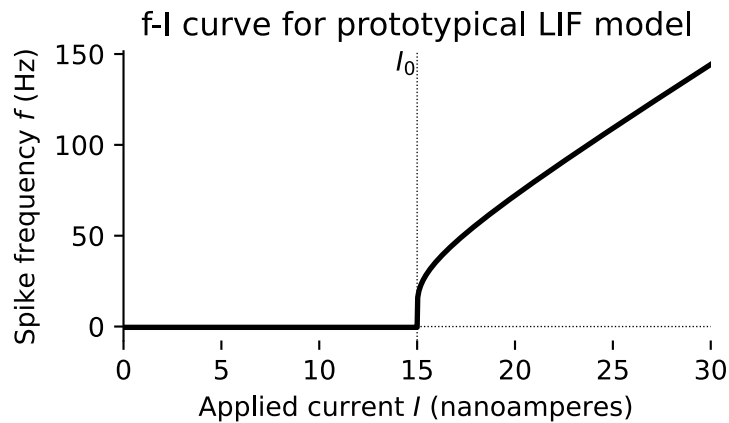
$$v(t) = e^{-t/\tau}(v(0) - v_\infty) + v_\infty. \quad (2)$$

and obtain the values of τ and v_∞ in terms of C , R , E_ℓ , and/or I . You may use any appropriate method (verifying by differentiating, method of undetermined coefficients, integrating factor, or methods not covered in this course such as Fourier or Laplace transforms). [6]

The most direct route is differentiate and match coefficients:

$$\begin{aligned}
 \dot{v}(t) &= \frac{d}{dt} \{ e^{-t/\tau} (v(0) - v_\infty) + v_\infty \} \\
 &= -\frac{1}{\tau} \underbrace{e^{-t/\tau} (v(0) - v_\infty)}_{v(t) - v_\infty} \\
 &= -\frac{1}{\tau} (v(t) - v_\infty) \\
 &= \frac{1}{\tau} (v_\infty - v(t)) \\
 \underbrace{\tau}_{RC} \dot{v}(t) &= \underbrace{v_\infty}_{E_\ell + IR} - v(t)
 \end{aligned} \tag{3}$$

- iii.α Below is a plot of firing frequency as a function of injected current. The LIF model has a hard threshold: The model abruptly starts spiking above a certain minimum current (labelled I_0 below). Derive an expression for I_0 in terms of v_∞ , C , R . [5]



Note: You do not need to know the values of the model parameters here. No calculation is required and you may leave your answer in terms of v_∞ , C , R .

The neuron can only spike when $v_\infty > v_{\text{threshold}}$, and we find the point just before this by solving for $v_\infty = v_{\text{threshold}}$. Depending on specific reading of the question, the following answers are valid

$$\begin{aligned}
 I_0 &> \frac{v_{\text{threshold}} - E_\ell}{R} \\
 I_0 &\geq \frac{v_{\text{threshold}} - E_\ell}{R} \\
 I_0 &= \frac{v_{\text{threshold}} - E_\ell}{R}
 \end{aligned} \tag{4}$$

Full marks for *any correct* nontrivial expression for I_0 . Partial marks for valiant attempts with hasty errors; -1 mark for a sign error or writing $<$ in equation (4).

- iii.β State whether you'd expect I_0 to increase or decrease if E_ℓ is *decreased*. Would this make it easier or harder for incoming excitatory synapses to cause this neuron to fire? Briefly speculate about what such a change might correspond to in a binary perceptron, whose output is given by $y = \Theta(\mathbf{w}^\top \mathbf{x} - b)$. (≈ 10 – 20 words) [5]

Students may refer to their solution in (iii.α) to note that decreasing the leak (resting) voltage E_ℓ increases the firing threshold. This has nothing to do with the **reset** voltage and marks will be lost for mentioning v_{reset} incorrectly. Alternatively, students may note that pulling the leak current lower will decrease v_∞ from part (ii), making some previously above-threshold currents fail to spike, i.e. increasing the threshold required to emit a spike. Alternatively, if neither of these solutions are available, students may intuit that E_ℓ “pulls the voltage down” in some sense (perhaps with a metaphor to an ideal linear spring), so lowering it further increases the amount of “force” (current) required to move the voltage a given displacement (toward threshold). The overall effect is quite similar to increasing the threshold b in the provided expression for the binary perceptron.

(c) [22 marks] The Hodgkin–Huxley Model: (10–20 minutes)

The voltage in the Hodgkin Huxley model of the action potential evolves according to the following differential equations:

$$\begin{aligned} C\dot{v} &= g_\ell(E_\ell - v) + \bar{g}_K n^4(E_K - v) + \bar{g}_{\text{Na}} m^3 h(E_{\text{Na}} - v) + I \\ \dot{n} &= \alpha_n(v) \cdot (1 - n) - \beta_n(v) \cdot n \\ \dot{m} &= \alpha_m(v) \cdot (1 - m) - \beta_m(v) \cdot m \\ \dot{h} &= \alpha_h(v) \cdot (1 - h) - \beta_h(v) \cdot h. \end{aligned} \quad (5)$$

You do not need to know the specific form of the nonlinear functions $\alpha_{\dots}(v)$ and $\beta_{\dots}(v)$ for this question.

- i. What is the physiological interpretation of the constants $\bar{g}_\ell$, $\bar{g}_K$, $\bar{g}_{\text{Na}}$? Provide physical units for these variables and discuss their role in the voltage dynamics. (≈ 25 – 50 words) [2]

These are the maximal conductances, and have units of inverse Ohms or Siemens (sometimes expressed also as per unit area, although other units would need to change or conversion terms added for this to make sense here). They determine the maximal conductance of the various ion channels, in the scenario where all ion channels are open and not inactivated, i.e. all ion channels are currently available to conduct current. (The maximal conductances are important as adding/removing ion channels is one robust way for neurons to homeostatically tune and adjust their electronic properties, although this level of depth is not required).

- ii. What part of the action potential does the gating variable h on the sodium conductance capture? (≈ 25 – 50 words) [2]

The h gate captures sodium channel inactivation. During the action potential, this stops the inward flow of positive sodium ions, limits depolarisation, and allows voltage-gated potassium channels to repolarise the membrane. Inactivation of sodium channels is also critical to the absolute refractory period, during which time there is not enough sodium conductance to initiate an action potential. (De-inactivation of voltage-gated sodium is also important for rebound spiking, but we did not cover this). There is some fungibility of the specific voltage h kicks in here, as some models will have inactivation begin at relatively low voltages (but kick in with some delay, allowing brief depolarisation), and others may have inactivation kick in faster at a slightly higher threshold.

- iii. **Voltage clamp:** Recall that the functions $\alpha_{\dots}(v)$ and $\beta_{\dots}(v)$ can be experimentally measured by using a feedback amplifier to inject current into the neuron to hold $v(t)$ at a constant value, thereby setting $\alpha_{\dots}$ and $\beta_{\dots}$ to constant values, and turning the ODEs for gating variables n, m, h into stable first-order linear differential equations. This is called a “voltage clamp” recording. Throughout this sub-question, assume that voltage v and gating variables $\alpha_{\dots}$ and $\beta_{\dots}$ are unknown constants, and that the linear ODEs for the gating variables are stable.

- iii.a For a given constant voltage v , derive an expression for the steady-state value for the gating variable h , denoted as $h_{\infty} = \lim_{t \rightarrow \infty} h(t)$ in terms of $\alpha_h(v)$ and $\beta_h(v)$. [6]

Setting $\dot{h}$ to zero gives $\alpha(1 - h) = \beta h$. Solving for h yields $\alpha/(\alpha + \beta)$, which is a fixed point of $\dot{h}$. As this equation is stable, this fixed point is the steady-state voltage $h_{\infty} = \alpha/(\alpha + \beta)$.

- iii.b Re-write the differential equation for the sodium gating variable in the form

$$\tau_h(v)\dot{h} = h_{\infty}(v) - h \quad (6)$$

and give an expression for $\tau_h(v)$ in terms of $\alpha_h(v)$ and $\beta_h(v)$. [6]

$$\begin{aligned} \dot{h} &= \alpha \cdot (1 - h) - \beta h \\ \dot{h} &= \alpha - \alpha h - \beta h \\ \dot{h} &= \alpha - (\alpha + \beta) \cdot h \\ \frac{1}{\alpha + \beta} \dot{h} &= \frac{\alpha}{\alpha + \beta} - h \\ \text{let } \tau_h &:= \frac{1}{\alpha + \beta} \\ \text{let } h_{\infty} &:= \frac{\alpha}{\alpha + \beta} \\ \text{then } \tau_h \dot{h} &= h_{\infty} - h \end{aligned} \quad (7)$$

Since this question involves transforming two given expressions into each other, showing work is required. Only partial marks can be given for answers that try to work from both ends but show errors or vagueness in the middle. Correct expressions for τ_h and h_{∞} without incorrect mathematics hint at rote memorisation, but these answers will still receive some marks.

- iii.c** Pretend that you have measured $\tau_h(v)$ and $h_\infty(v)$ from a voltage-clamp recording at voltage v . Derive expressions to calculate $\alpha_h(v)$ and $\beta_h(v)$ from $\tau_h(v)$ and $h_\infty(v)$. **[6]**

We need to algebraically solve for α and β from the expressions for τ_h and h_∞ in parts (a) and (b). Here is one way: Nothing that

$$\begin{aligned}\tau_h &= \frac{1}{\alpha + \beta} \\ h_\infty &= \frac{\alpha}{\alpha + \beta},\end{aligned}\tag{8}$$

we have that $h_\infty = \alpha\tau_h$ and solving for α gives $\alpha = h_\infty/\tau_h$. We can then substitute this in to the expression for τ_h to recover $\beta = \tau_h^{-1} - \alpha = (1 - h_\infty)/\tau_h$.

Any correct solution that shows key steps, assuring the markers that the answer has simply not been memorised, will be valid. This question can also be answered by asserting the solution, and showing via algebraic manipulation or other methods of proof that it is correct (reduces to already given or proven identities). If work in previous parts of the question is used to support this, this work should be referenced explicitly and clearly.